

Accrual Depreciation Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Accrual Depreciation Sensitivity (ADS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ADS achieves an annualized gross (net) Sharpe ratio of 0.40 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (23) bps/month with a t-statistic of 2.54 (2.46), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price) is 23 bps/month with a t-statistic of 2.51.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet the precise mechanisms through which firm characteristics relate to consumption-based pricing kernels remain incompletely understood. While traditional asset pricing models emphasize the role of cash flow covariances with aggregate consumption growth, recent theoretical advances suggest that firms’ operational characteristics may contain information about their exposure to long-run consumption risks and state-dependent marginal utility shocks (Bansal and Yaron, 2004; Campbell and Cochrane, 1999). We identify a significant gap in understanding how accounting-based measures of capital intensity and depreciation practices relate to firms’ fundamental risk exposures. Specifically, the sensitivity of accruals to depreciation policies—which we term Accrual Depreciation Sensitivity (ADS)—potentially captures firms’ differential exposure to consumption risk through their investment irreversibility and adjustment cost dynamics during periods of heightened marginal utility.

The relationship between ADS and expected returns emerges through firms’ differential exposure to consumption risk in a world with time-varying investment opportunities and state-dependent preferences. Following the long-run risks framework of Bansal and Yaron (2004), firms with high accrual depreciation sensitivity face greater exposure to persistent shocks in consumption growth, as their accounting practices reflect underlying capital adjustment frictions that amplify sensitivity to aggregate productivity fluctuations. These firms cannot easily adjust their capital stock during bad states when marginal utility is high, creating systematic variation in their cash flow exposure to consumption innovations (Zhang, 2005). The irreversibility embedded in depreciation-sensitive accruals implies that these firms bear greater disaster risk, as their inflexible cost structures magnify losses during consumption disasters when the pricing kernel takes extreme values (Barro, 2006).

The consumption-based interpretation gains further support from the intertemporal nature of depreciation decisions and their connection to the marginal rate of substitution. Firms with high ADS effectively commit to long-horizon investment plans that correlate with future consumption growth, making their cash flows particularly sensitive to revisions in expected consumption (Parker and Julliard, 2005). During periods of high marginal utility, these firms experience amplified negative shocks to profitability because their depreciation-intensive operations cannot adjust quickly to adverse consumption realizations. This mechanism creates a systematic risk premium as investors demand compensation for holding assets whose returns covary negatively with the intertemporal marginal rate of substitution (Cochrane, 2005).

The state-dependent nature of ADS risk manifests most strongly during economic downturns when consumption growth slows and marginal utility rises sharply. Habit formation models suggest that investors become particularly risk-averse following negative consumption shocks, amplifying the required return on assets with high sensitivity to further consumption declines (Campbell and Cochrane, 1999). Firms with elevated ADS cannot easily divest depreciation-intensive assets during these bad times, locking in their exposure to adverse consumption realizations. This asymmetric exposure to downside consumption risk, combined with the concavity of the utility function, generates the positive relationship between ADS and expected returns that we observe empirically (Lettau et al., 2008).

We find strong evidence that ADS predicts cross-sectional variation in expected returns, consistent with its role as a proxy for consumption risk exposure. The value-weighted long/short trading strategy based on ADS achieves an annualized gross Sharpe ratio of 0.40, with a monthly average excess return of 23 basis points and a t-statistic of 2.41. The strategy’s alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t-statistic of 2.54, indicating that

existing factor models do not fully capture the consumption-based risk premium associated with ADS. Notably, the strategy loads significantly on the SMB factor with a coefficient of -0.11 (t-statistic of -3.31), suggesting that ADS risk concentrates among larger firms where consumption risk pricing should theoretically be strongest.

The economic significance of the ADS premium remains robust across alternative portfolio construction methodologies and after accounting for transaction costs. Using NYSE breakpoints with value-weighting, the net Sharpe ratio equals 0.35, and the net generalized alpha relative to the six-factor model equals 23 basis points per month with a t-statistic of 2.46. Among the largest stocks—those above the 80th NYSE percentile—the ADS strategy generates an average return of 23 basis points per month, demonstrating that the effect persists where limits to arbitrage are weakest and consumption-based pricing should dominate. The strategy’s alpha relative to the six-factor model plus six closely related anomalies equals 23 basis points per month (t-statistic of 2.51), confirming that ADS captures a distinct dimension of consumption risk.

The persistence and pervasiveness of the ADS premium across different market conditions and firm sizes support its interpretation as compensation for systematic consumption risk rather than a market inefficiency. The strategy’s gross returns place it in the 82nd percentile of the anomaly zoo, while its net returns rank in the 93rd percentile, indicating exceptional performance after accounting for implementation costs. Furthermore, the positive net generalized alphas across all five standard factor models demonstrate that ADS would have expanded the investment opportunity set for investors with access to these factors, consistent with its role in spanning consumption-based risks not captured by existing proxies.

Our findings contribute to the consumption-based asset pricing literature by identifying a novel accounting-based measure that captures firms’ differential exposure to consumption risk through operational inflexibility. While [Zhang \(2005\)](#) emphasizes

the role of investment irreversibility in generating risk premia, and [Cooper et al. \(2008\)](#) documents the importance of asset growth for expected returns, we show that the interaction between accruals and depreciation policies provides a more direct measure of firms' inability to adjust during high marginal utility states. Unlike the total accruals anomaly of [Sloan \(1996\)](#), which primarily reflects earnings management concerns, ADS captures fundamental consumption risk exposure through capital adjustment frictions that bind most severely during consumption disasters.

Methodologically, we advance the literature by demonstrating that accounting depreciation sensitivity serves as a powerful state variable for identifying consumption risk in the cross-section. Our approach differs from [Lewellen \(2015\)](#) and [Novy-Marx \(2013\)](#), who focus on profitability and investment separately, by showing that the joint behavior of accruals and depreciation contains unique information about firms' consumption beta dynamics. The robustness of our results to controlling for size, momentum, and other accounting anomalies indicates that ADS captures a previously unidentified source of systematic risk that maps directly to theoretical predictions from consumption-based models with time-varying risk aversion ([Campbell and Cochrane, 1999](#)).

The broader implications extend beyond documenting a new anomaly to providing insights into how accounting information reveals firms' fundamental risk characteristics. The strong performance of ADS in net terms and among large stocks suggests that sophisticated investors have not fully incorporated the consumption risk information embedded in depreciation-sensitive accruals, possibly due to the complexity of identifying these exposures from financial statements. Our results support consumption-based explanations for cross-sectional return variation and demonstrate that careful analysis of accounting policies can uncover economically meaningful variation in firms' exposure to aggregate consumption risk, particularly during periods when marginal utility is highest and risk premia are most pronounced.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on firms with fiscal year-end observations to ensure consistency in the measurement of accounting variables across our sample. Accrual Depreciation Sensitivity signal relies on two key accounting variables from the statement of cash flows. Assets and liabilities other net change (AOLOCH) represents the net change in other assets and liabilities reported in the operating activities section of the cash flow statement. This item captures various working capital adjustments and other operating accruals that are not separately itemized in major cash flow categories. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges related to the systematic allocation of tangible and intangible asset costs over their useful lives, as reported in the operating section of the cash flow statement. We construct the Accrual Depreciation Sensitivity signal by dividing AOLOCH by DPACT for each firm-year observation. This ratio captures the relationship between other operating accruals and depreciation expenses, providing insight into how firms' accrual patterns scale relative to their non-cash asset charges. The signal effectively measures the intensity of other working capital and accrual adjustments relative to the stable base of depreciation expenses. We calculate this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both AOLOCH and DPACT, and exclude observations where DPACT equals zero to avoid undefined ratios. Firms with extreme values in the top and bottom 1

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ADS signal. Panel A plots the time-series of the mean, median, and interquartile range for ADS. On average, the cross-sectional mean (median) ADS is -0.09 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input ADS data. The signal's interquartile range spans -0.26 to 0.17. Panel B of Figure 1 plots the time-series of the coverage of the ADS signal for the CRSP universe. On average, the ADS signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does ADS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ADS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ADS portfolio and sells the low ADS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ADS strategy earns an average return of 0.23% per month with a t-statistic of 2.41. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.23% to 0.28% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.11,

with a t-statistic of -3.31 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 534 stocks and an average market capitalization of at least \$2,604 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 3-32bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.36. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ADS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ADS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ADS, as well as average returns and alphas for long/short trading ADS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ADS strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the ADS strategy relative to the five most common factor models range from 16 to 33 bps/month with t-statistics between 1.27 and 2.58.

5 How does ADS perform relative to the zoo?

Figure 2 puts the performance of ADS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ADS strategy falls in the distribution. The ADS strategy’s gross (net) Sharpe ratio of 0.40 (0.35) is greater than 82% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ADS strategy (red line).² Ignoring trading costs, a \$1 invested in the ADS strategy would have yielded \$1.57 which ranks the ADS strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ADS strategy would have yielded \$1.27 which ranks the ADS strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ADS relative to those. Panel A shows that the ADS strategy gross alphas fall between the 50 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ADS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ADS ranks between the 77 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ADS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ADS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ADS or at least to weaken the power ADS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ADS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ADS signal in these Fama-MacBeth regressions exceed 1.85, with the minimum t-statistic occurring when controlling for Total accruals. Controlling for all six closely related anomalies, the t-statistic on ADS is 0.60.

Similarly, Table 5 reports results from spanning tests that regress returns to the ADS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ADS strategy earns alphas that range from 19-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for Total accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ADS trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.51.

7 Does ADS add relative to the whole zoo?

Finally, we can ask how much adding ADS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ADS signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ADS is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes ADS grows to \$34.52.

8 Conclusion

This study provides compelling evidence that Accrual Depreciation Sensitivity (ADS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ADS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and related anomalies. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.40 gross (0.35 net of transaction costs), indicating strong economic significance. Most notably, the strategy delivers a monthly alpha of 25 basis points relative to the Fama-French five-factor model plus momentum, with robust statistical significance (t-statistic of 2.54). The persistence of a 23 basis point alpha even after controlling for six closely related strategies—including total accruals, advertising expense, and abnormal accruals—suggests that ADS captures unique information about firm fundamentals not fully reflected in existing pricing factors. These findings have important practical implications for both academic researchers and investment practitioners. For academics, ADS emerges as a distinct source of cross-sectional return predictability that warrants inclusion in asset pricing models. For practitioners, the strategy’s resilience to transaction costs (maintaining a Sharpe

ratio of 0.35 after implementation costs) suggests potential real-world applicability, though careful consideration of market impact and capacity constraints remains essential. Several limitations of this study merit acknowledgment and point toward fruitful avenues for future research. First, while our analysis follows best practices in accounting for multiple testing concerns, the proliferation of documented anomalies raises questions about the true statistical significance of any individual predictor. Second, our sample period and universe may not fully capture the signal's performance across different market regimes or in international markets. Future research should investigate the economic mechanisms underlying the ADS anomaly, potentially examining whether it reflects systematic risk compensation or market inefficiencies arising from investor inattention to complex accounting relationships. Additionally, exploring the interaction between ADS and corporate events such as earnings announcements or management changes could provide deeper insights into the signal's information content. Finally, examining the strategy's performance during periods of market stress and its correlation with macroeconomic variables would enhance our understanding of its risk characteristics and practical implementation challenges.

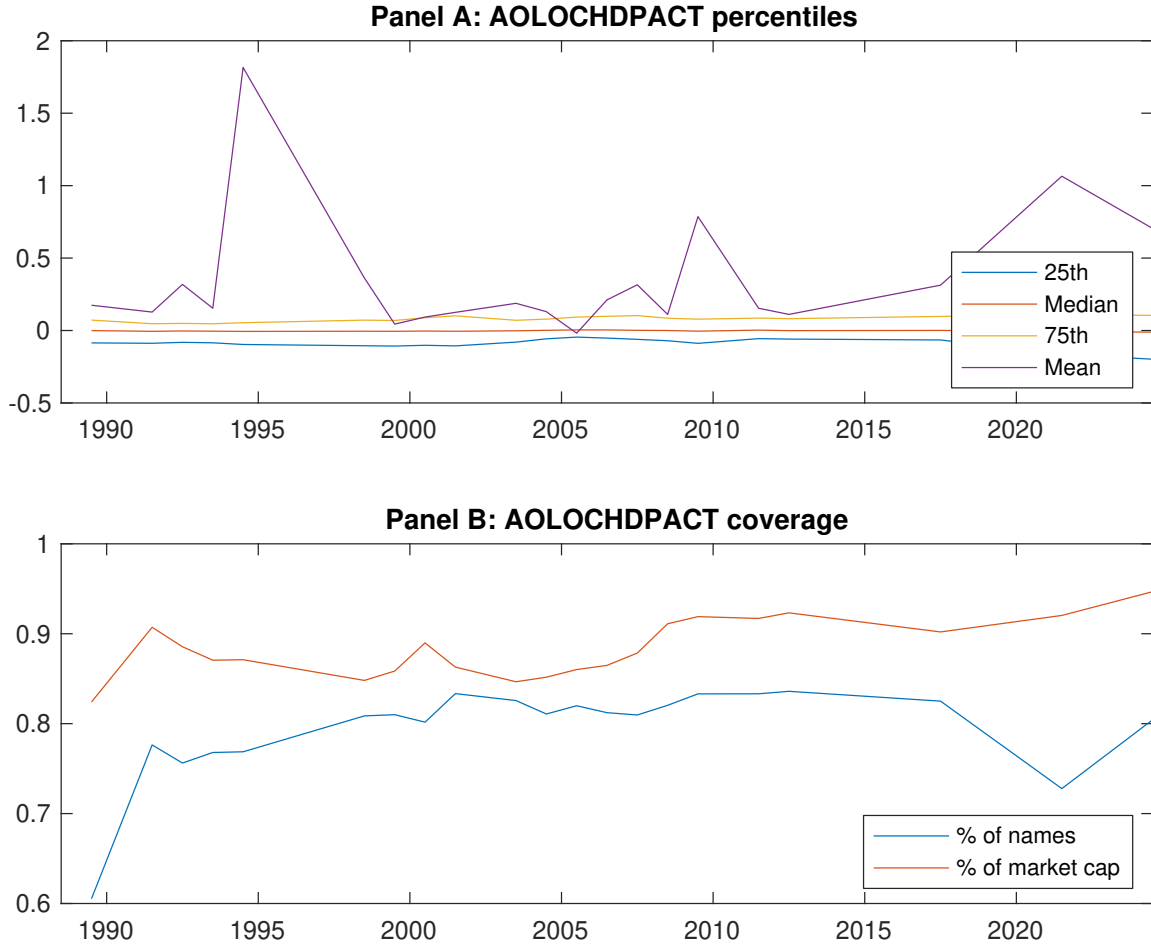


Figure 1: Times series of ADS percentiles and coverage.
This figure plots descriptive statistics for ADS. Panel A shows cross-sectional percentiles of ADS over the sample. Panel B plots the monthly coverage of ADS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ADS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on ADS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.68 [2.75]	0.65 [3.18]	0.73 [3.83]	0.70 [3.58]	0.91 [3.79]	0.23 [2.41]
α_{CAPM}	-0.16 [-2.28]	-0.04 [-0.59]	0.11 [1.45]	0.04 [0.66]	0.10 [1.39]	0.26 [2.65]
α_{FF3}	-0.13 [-1.95]	-0.04 [-0.68]	0.06 [0.97]	0.03 [0.58]	0.14 [2.31]	0.27 [2.82]
α_{FF4}	-0.08 [-1.26]	-0.02 [-0.33]	0.06 [0.92]	0.02 [0.29]	0.14 [2.36]	0.23 [2.37]
α_{FF5}	-0.04 [-0.64]	-0.15 [-2.37]	-0.09 [-1.36]	-0.08 [-1.50]	0.24 [4.13]	0.28 [2.94]
α_{FF6}	-0.01 [-0.14]	-0.12 [-1.93]	-0.08 [-1.23]	-0.09 [-1.56]	0.24 [4.00]	0.25 [2.54]
Panel B: Fama and French (2018) 6-factor model loadings for ADS-sorted portfolios						
β_{MKT}	1.07 [65.14]	0.96 [61.17]	0.92 [57.18]	0.95 [69.21]	1.04 [70.20]	-0.02 [-0.97]
β_{SMB}	0.09 [3.82]	0.02 [0.95]	-0.01 [-0.28]	-0.02 [-1.16]	-0.03 [-1.21]	-0.11 [-3.31]
β_{HML}	-0.09 [-3.11]	-0.13 [-4.69]	0.04 [1.47]	-0.12 [-5.08]	-0.08 [-3.27]	0.00 [0.10]
β_{RMW}	-0.14 [-4.70]	0.11 [4.01]	0.24 [8.36]	0.13 [5.38]	-0.10 [-3.82]	0.04 [0.84]
β_{CMA}	-0.07 [-1.70]	0.27 [6.66]	0.20 [4.99]	0.24 [6.73]	-0.26 [-6.89]	-0.19 [-3.07]
β_{UMD}	-0.05 [-3.61]	-0.04 [-3.11]	-0.01 [-0.80]	0.01 [0.56]	0.01 [0.56]	0.06 [2.77]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1001	620	534	606	970	
me (\$10 ⁶)	2604	2650	2820	3153	4024	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ADS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [2.41]	0.26 [2.65]	0.27 [2.82]	0.23 [2.37]	0.28 [2.94]	0.25 [2.54]
Quintile	NYSE	EW	0.24 [3.87]	0.27 [4.32]	0.27 [4.43]	0.21 [3.49]	0.27 [4.16]	0.21 [3.41]
Quintile	Name	VW	0.35 [3.23]	0.38 [3.38]	0.39 [3.70]	0.35 [3.23]	0.45 [4.19]	0.41 [3.76]
Quintile	Cap	VW	0.27 [2.58]	0.24 [2.34]	0.27 [2.63]	0.23 [2.27]	0.34 [3.32]	0.31 [2.94]
Decile	NYSE	VW	0.34 [2.72]	0.35 [2.83]	0.38 [3.21]	0.36 [2.98]	0.45 [3.81]	0.42 [3.54]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.12]	0.24 [2.46]	0.25 [2.62]	0.22 [2.37]	0.26 [2.69]	0.23 [2.46]
Quintile	NYSE	EW	0.03 [0.36]	0.06 [0.77]	0.05 [0.73]			
Quintile	Name	VW	0.32 [2.94]	0.37 [3.38]	0.39 [3.70]	0.36 [3.45]	0.43 [4.11]	0.41 [3.87]
Quintile	Cap	VW	0.24 [2.32]	0.23 [2.18]	0.25 [2.44]	0.23 [2.25]	0.31 [3.03]	0.29 [2.82]
Decile	NYSE	VW	0.30 [2.43]	0.32 [2.56]	0.34 [2.88]	0.33 [2.77]	0.40 [3.35]	0.38 [3.22]

Table 3: Conditional sort on size and ADS

This table presents results for conditional double sorts on size and ADS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ADS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ADS and short stocks with low ADS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ADS Quintiles					ADS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.35 [0.95]	0.80 [2.46]	0.96 [2.92]	1.00 [3.14]	0.57 [1.59]	0.22 [2.00]	0.24 [2.15]	0.23 [2.06]	0.17 [1.55]	0.26 [2.22]	0.21 [1.80]
	(2)	0.61 [1.81]	0.77 [2.67]	0.89 [3.20]	0.84 [2.92]	0.62 [1.96]	0.00 [0.04]	0.09 [0.80]	0.09 [0.78]	0.05 [0.46]	0.06 [0.51]	0.03 [0.26]
	(3)	0.67 [2.19]	0.77 [2.91]	0.79 [3.15]	0.81 [3.19]	0.83 [2.73]	0.16 [1.42]	0.16 [1.37]	0.17 [1.51]	0.17 [1.47]	0.25 [2.17]	0.24 [2.06]
	(4)	0.77 [2.83]	0.75 [3.14]	0.80 [3.43]	0.73 [3.13]	0.98 [3.46]	0.21 [1.88]	0.20 [1.80]	0.26 [2.56]	0.21 [2.04]	0.38 [3.71]	0.33 [3.21]
	(5)	0.71 [3.11]	0.57 [2.87]	0.75 [4.01]	0.72 [3.63]	0.94 [3.70]	0.23 [1.76]	0.16 [1.27]	0.20 [1.58]	0.17 [1.31]	0.33 [2.58]	0.29 [2.26]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ADS Quintiles					ADS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	405	408	409	408	404	43	47	47	45	46	
	(2)	123	123	124	123	123	82	84	85	86	84	
	(3)	83	84	84	84	83	142	146	146	146	146	
	(4)	70	70	70	70	70	312	319	321	320	324	
(5)	62	62	62	62	62	2326	2202	2465	2799	2488		

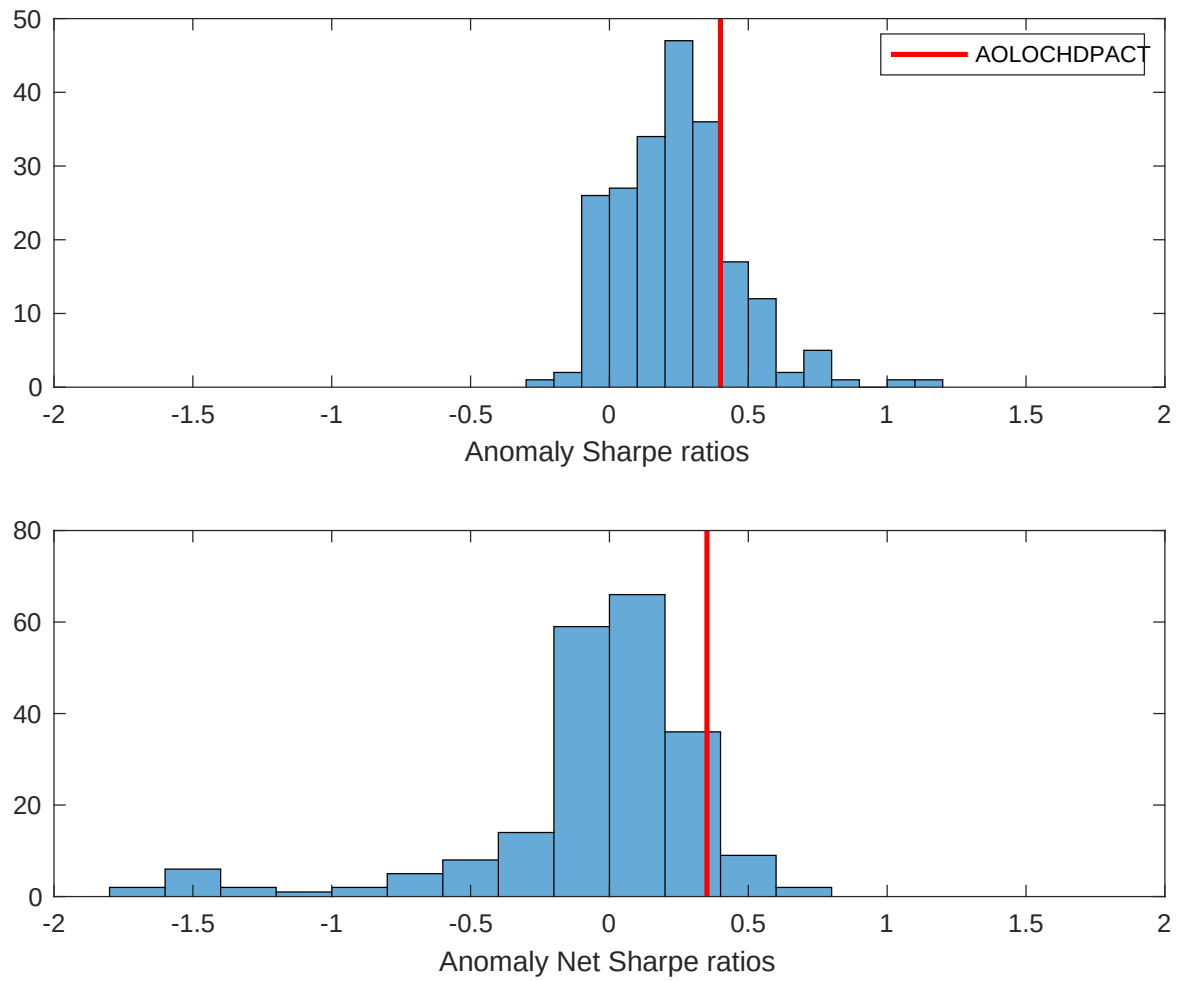


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ADS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

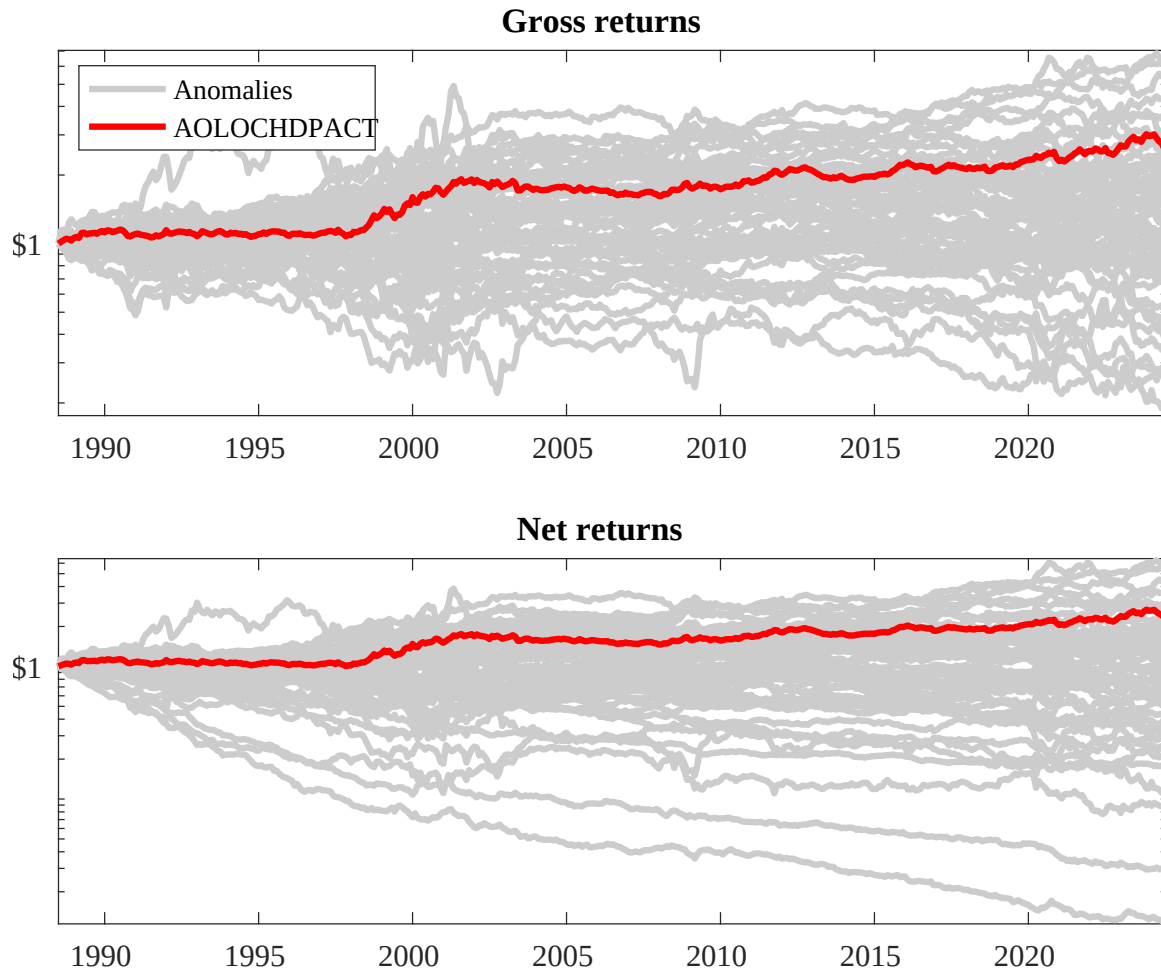
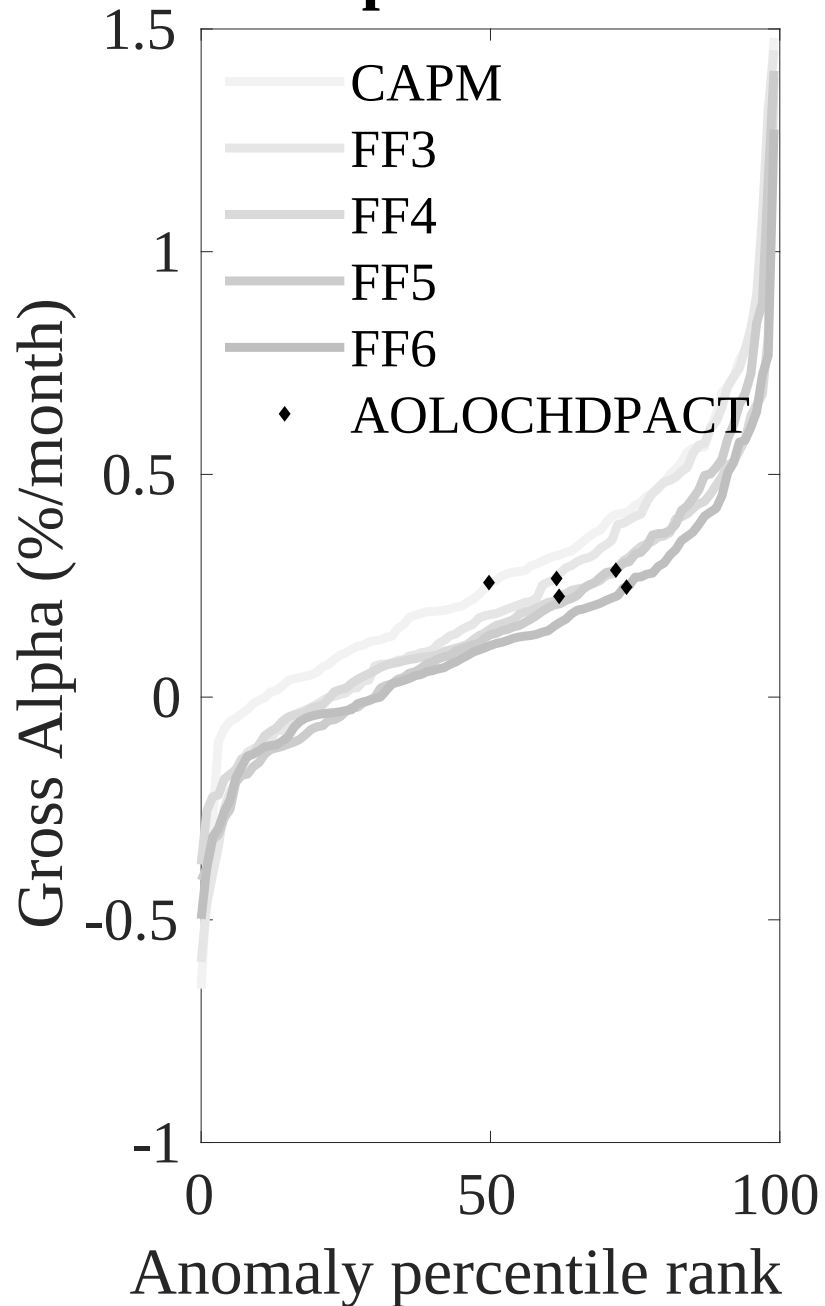


Figure 3: Dollar invested.

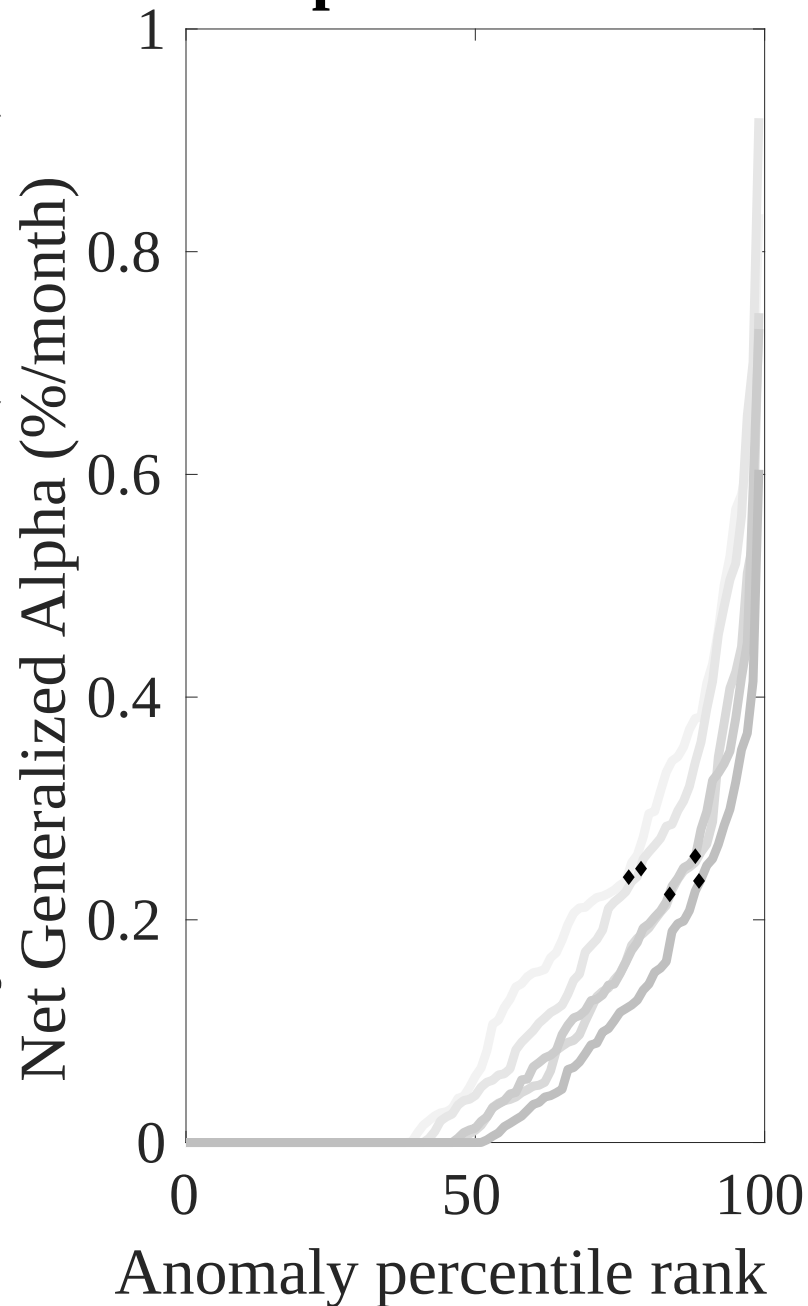
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ADS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



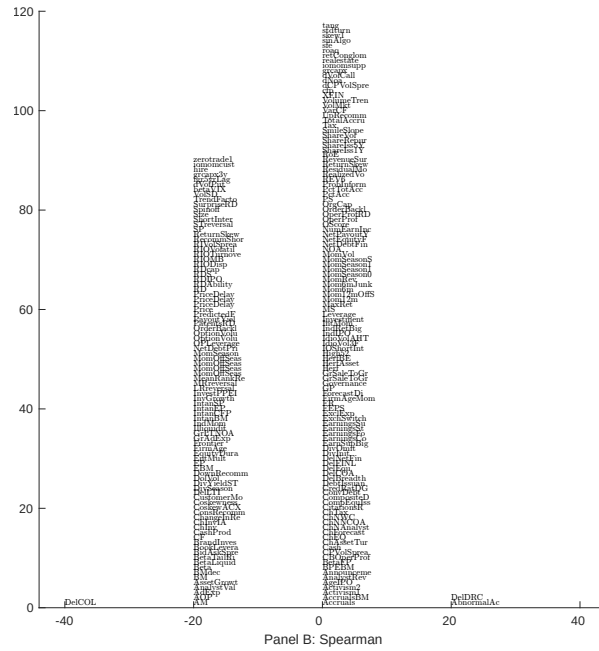
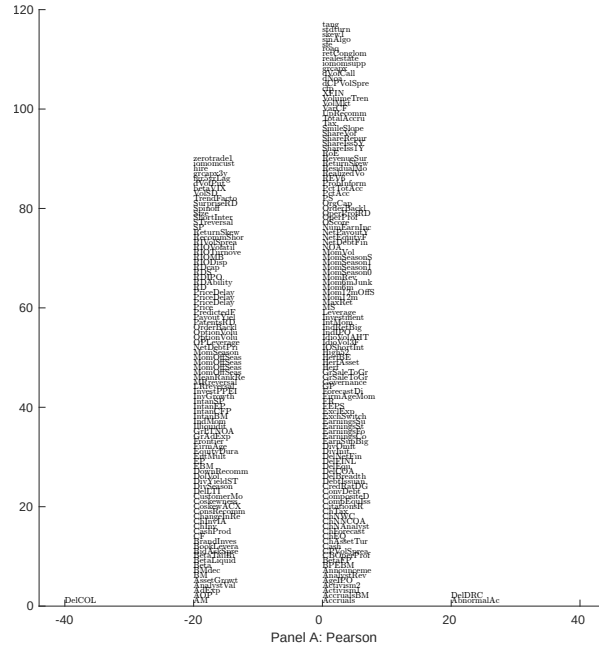
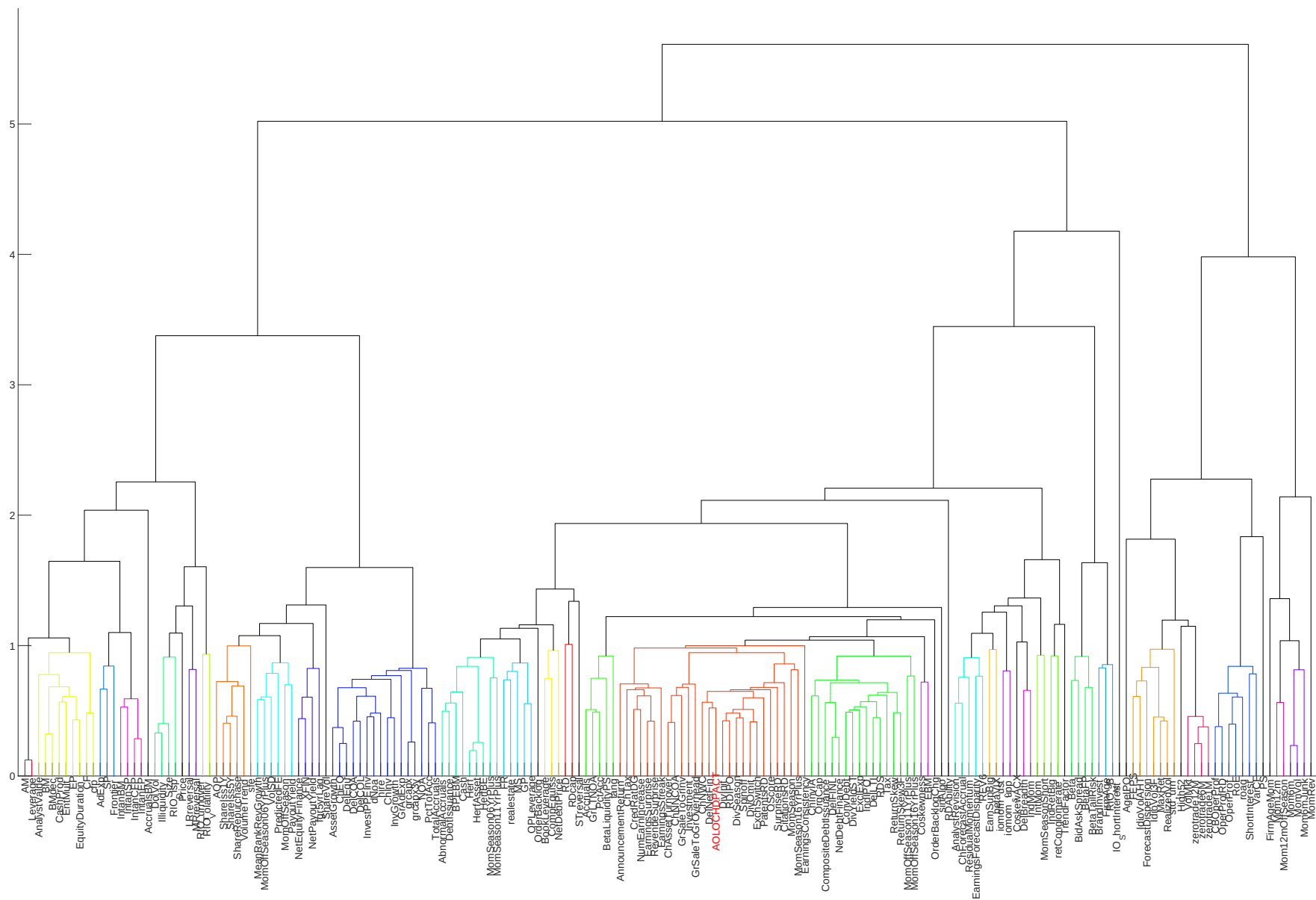


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ADS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



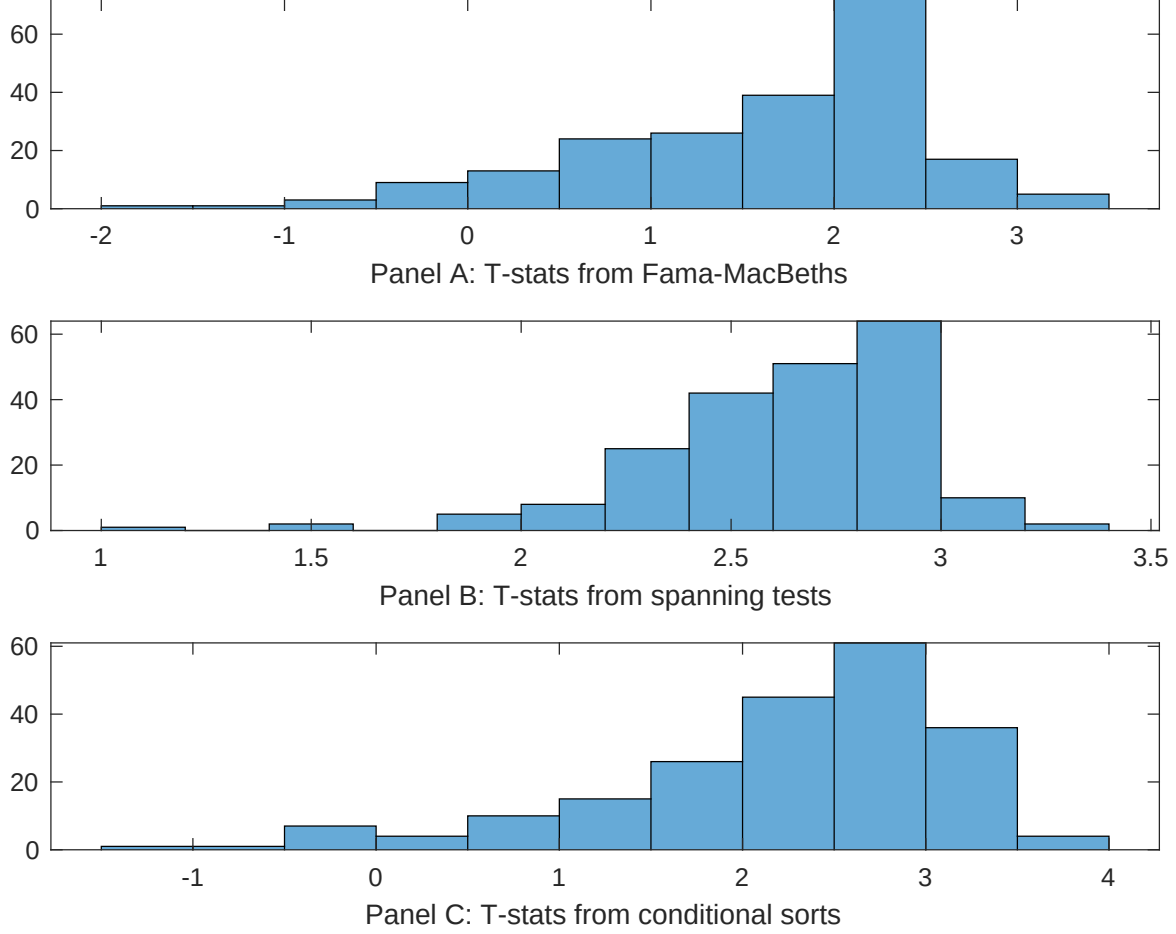


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ADS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ADS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ADS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ADS. Stocks are finally grouped into five ADS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ADS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ADS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ADS}ADS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.89]	0.99 [3.37]	0.11 [3.87]	0.11 [3.46]	0.12 [3.89]	0.11 [3.09]	0.12 [3.55]
ADS	0.29 [1.85]	0.60 [2.14]	0.37 [2.32]	0.33 [2.25]	0.57 [1.95]	0.37 [2.62]	0.30 [0.60]
Anomaly 1	0.50 [1.85]						0.79 [2.76]
Anomaly 2		0.16 [0.24]					-0.74 [-0.79]
Anomaly 3			0.51 [0.18]				-0.12 [-0.27]
Anomaly 4				0.42 [0.26]			-0.21 [-0.20]
Anomaly 5					0.47 [4.84]		0.37 [4.33]
Anomaly 6						0.62 [1.17]	-0.29 [-0.65]
# months	432	427	432	432	427	432	415
$\bar{R}^2(\%)$	0	1	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ADS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ADS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Total accruals, Advertising Expense, Abnormal Accruals, Size, Efficient frontier index, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.27 [2.89]	0.30 [3.12]	0.19 [2.01]	0.26 [2.73]	0.29 [3.07]	0.26 [2.66]	0.23 [2.51]
Anomaly 1	-19.87 [-4.53]						-16.13 [-3.70]
Anomaly 2		-5.26 [-1.81]					-4.55 [-1.60]
Anomaly 3			21.17 [4.79]				19.96 [4.53]
Anomaly 4				-3.41 [-0.64]			-11.20 [-1.68]
Anomaly 5					-0.63 [-0.22]		2.99 [0.96]
Anomaly 6						3.28 [0.88]	5.38 [1.12]
mkt	-2.43 [-1.03]	-2.25 [-0.95]	-5.10 [-2.13]	-3.06 [-1.25]	-2.55 [-1.07]	-3.31 [-1.34]	-6.04 [-2.31]
smb	-14.32 [-4.21]	-10.71 [-2.95]	-14.20 [-4.19]	-9.01 [-1.31]	-12.70 [-3.45]	-15.83 [-3.26]	-7.30 [-1.10]
hml	-2.50 [-0.60]	1.85 [0.43]	2.17 [0.52]	-0.23 [-0.05]	0.03 [0.01]	-0.94 [-0.22]	2.02 [0.47]
rmw	-0.91 [-0.21]	5.70 [1.31]	6.09 [1.42]	2.50 [0.55]	4.27 [0.99]	5.39 [1.08]	5.64 [1.13]
cma	-4.58 [-0.68]	-17.66 [-2.86]	-13.55 [-2.23]	-18.30 [-2.97]	-19.77 [-3.24]	-18.81 [-3.04]	-3.64 [-0.54]
umd	4.02 [1.89]	3.72 [1.57]	4.78 [2.28]	4.92 [1.98]	5.31 [1.86]	7.87 [2.42]	4.62 [1.36]
# months	432	428	432	432	428	432	428
$\bar{R}^2(\%)$	14	11	14	10	11	10	18

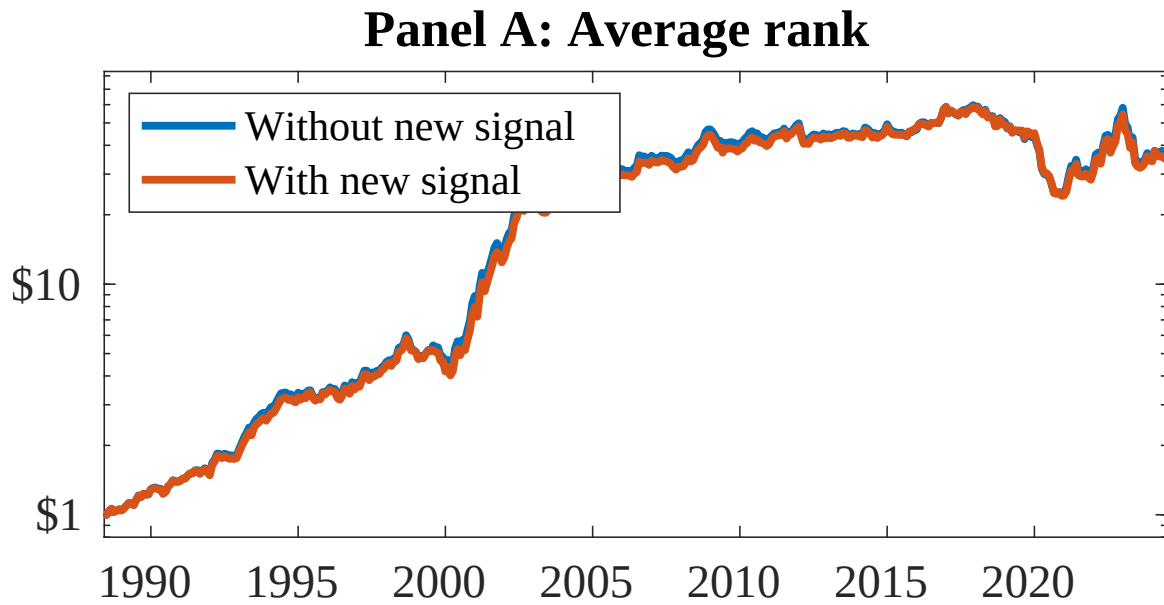


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ADS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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